

Credit Card Expenditure Predictions

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Introduction:

Understanding what drives an individual's credit card expenditure is important for financial institutions and people seeking to make informed decisions about their finances. In this project, we investigate the factors of credit card spending using the *Credit Card Data from the textbook Econometric Analysis*, dataset in Kaggle. The dataset provides detailed information on credit card holders such as share, dependents, major derogatory reports, age, and expenditure.

The central problem we address is how consumer demographics and credit histories can be used to predict monthly expenditure on a credit card. We aim to identify which factors contribute most strongly to monthly expenditure and whether certain combinations of variables play a meaningful role.

Our core research questions:

- 1) What are the main predictors of monthly credit card expenditure?
- 2) Does the interaction between age and income positively influence expenditure?
- 3) Do major derogatory credit reports significantly affect spending behavior?

By better understanding the characteristics that predict expenditure, consumers can gain insight into the financial behaviors associated with credit card usage.

We began with a comprehensive exploratory data analysis. Based on these results, we selected a subset of the most influential predictors, such as age, income, number of dependents, credit rating measures, and others. We then constructed multiple linear regression models in R to interpret the predictability on monthly expenditure. The analysis included checking key model assumptions, interpreting statistically significant coefficients, and comparing different model specifications in order to identify the strongest predictors.

Regression Analysis: Exploratory Data Analysis (EDA)

Our EDA revealed important patterns and relationships, particularly among the numerical variables. Expenditure, share, months, and number of active accounts were strongly right-skewed with a few extreme outliers, which were visible in visualizations and likely to influence modeling. Correlation analysis showed the strongest relationship between expenditure and share which had $r = 0.83$. There were moderate correlations for dependents, age, and months. Surprisingly, income and expenditure exhibited only a weak relationship.

Among categorical variables, homeowners tended to spend more than renters, while individuals with derogatory credit reports generally earned and spent less. The variable selfemp, indicating self-employment, covered only about 7% of the sample and is unlikely to have a major role in the predictive model.

Overall, all of our random splits produced the same conclusions, agreeing that share had the strongest correlation with expenditure, and that much of the data had a strong skew and outliers. Transformations would probably be needed to satisfy assumptions such as equal variance and normality. The exploratory data analysis provided an ample amount of data for us to decide what path to take next when implementing our transformations on our model.

Initial Model

$$\text{expenditure} \sim \text{income} + \text{share} + \text{dependents} + \text{months} + \text{active}$$

To begin, we used the 5 variables with the highest correlation to expenditure: income, share, dependents, months, and active. This produced a good model, with a training adjusted R^2 of 0.7962, a testing adjusted R^2 of 0.848, and a RMSE of 101.7. With high adjusted R^2 s and a relatively low RMSE, our initial model is already quite well at predicting expenditure.

The residual plots of this initial model are problematic, however, with a clear pattern in the residuals vs fitted plot and significant deviations from the Q-Q line. This shows that the normality and constant variance assumptions are failing.

Transformations

In an effort to better satisfy the linear regression assumptions, we applied several targeted transformations and model adjustments. We first used a Box–Cox transformation on expenditure to reduce skewness and bring the response variable closer to normality ($\lambda = 0.5$). Additionally, because the variable share was heavily right-skewed, we applied a square root transformation to stabilize its distribution and reduce the influence of extreme values.

To further address the curvature observed in the Residuals vs Fitted plot, we explored higher-order terms, including both interaction effects and squared predictors. These terms were introduced to capture potential nonlinear relationships that the linear model could not represent on its own. Among all higher-order terms considered, the interaction between share and dependents was the only one that proved statistically significant, indicating that the effect of share on expenditure depends on the number of dependents. This interaction meaningfully improved model fit and was therefore retained in the final model.

Simplifying Model

Before model selection, we cleaned the dataset by removing observations with 0 dependents, which contradicted the variable's definition of "1 + number of dependents." Additionally, we excluded cases where expenditure was zero, as these corresponded to

individuals not approved for the credit card, and removed three extreme outliers to improve model fit.

To evaluate whether additional predictors improved overall model fit, we conducted a series of partial F-tests comparing nested models. When the p-value from a partial F-test is below the chosen α level, the additional predictors are considered to contribute significantly to the model and should be retained.

We compared models that included the predictors income, months, active, and the $\sqrt{\text{share}} \times \text{dependents}$ interaction term to reduced versions of the model. The tests indicated that adding months or active did not significantly reduce the residual sum of squares (both $p > .05$), and therefore these variables did not meaningfully improve the model.

Although the partial F-test for income was statistically significant, we ultimately chose to remove it because share is defined as the ratio of monthly credit card expenditure to yearly income. Since share is directly related to income, including both variables introduced unnecessary variability in the residual plots and complicated interpretation, even though multicollinearity between them was not severe. Thus, income, months, and active were excluded from the final model because they did not improve model performance.

In contrast, the partial F-test for the interaction between $\sqrt{\text{share}}$ and dependents was highly significant ($F \approx 40.45$, $p < .001$), indicating that the interaction substantially improved model fit. Based on these results, our final model retained the significant predictors, $\sqrt{\text{share}}$, dependents, and their interaction, and removed all non-significant variables.

Final Model

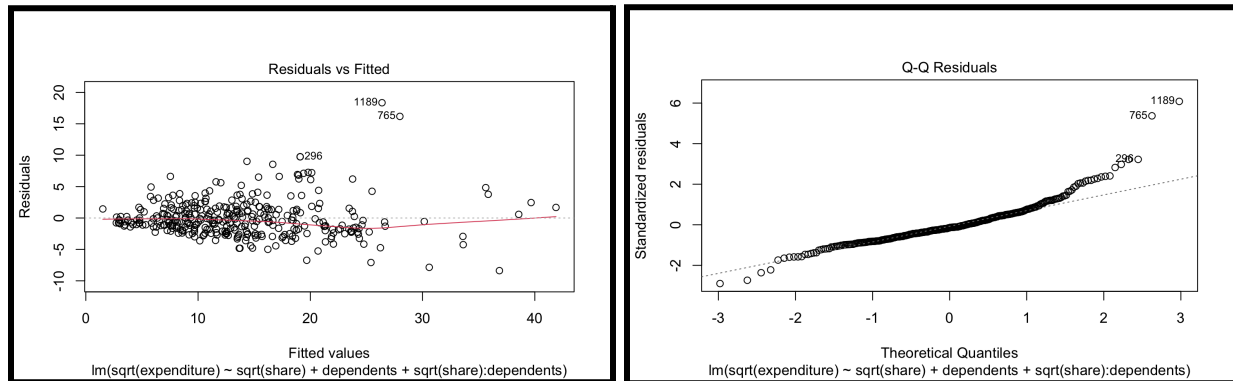
Our final model used a square-root transformation of the response variable, expenditure, to address the heteroskedasticity observed in the exploratory analysis and diagnostic plots. A Box–Cox analysis indicated that a transformation with $\lambda \approx 0.5$ was optimal, supporting the use of the square-root transformation. This approach substantially improved the residual patterns, reduced heteroskedasticity, and resulted in more stable variance across fitted values.

$$\sqrt{\text{expenditure}} = 1.3297 + 40.6148\sqrt{\beta_1} - 0.5161\beta_2 + 6.8362\sqrt{\beta_1(\beta_2)}$$

Where $\beta_1 = \text{share}$, $\beta_2 = \text{dependents}$

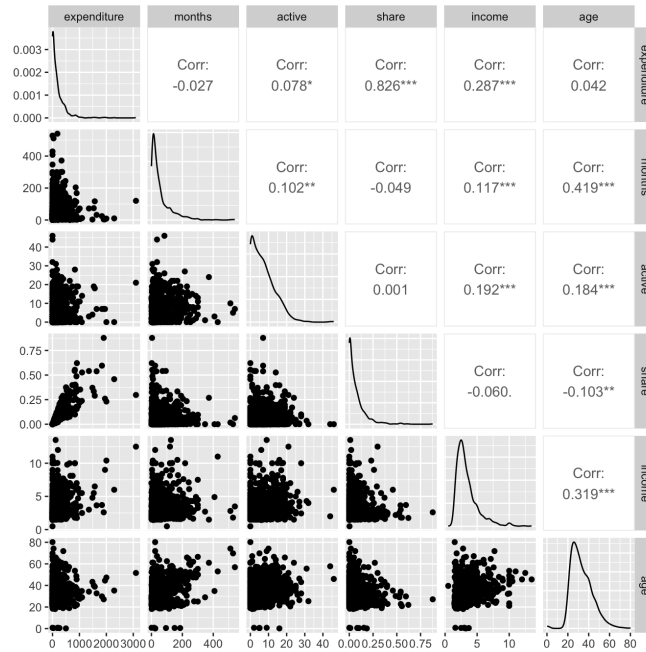
Our final model produced a training adjusted R^2 of 0.8391, a testing adjusted R^2 of 0.7153, and a RMSE of 4.34.

Residual Analysis:



While our transformations helped our residual analysis appear closer to passing, there is still a clear pattern in the residuals vs. fitted plot and noticeable deviations from the Q-Q line in the Q-Q plot. This suggests that our normality assumption and constant variance assumption are both still failing. We ran a Shapiro-Wilk's test and got a very small p-value, which confirms that our residuals are not normally distributed. Our Durbin-Watson test shows that the independence assumption is satisfied, however. Using our visual and statistical residual analysis, we can

conclude that our normality assumption and constant variance assumption fail, while our autocorrelation assumption passes.



The normality and heteroskedasticity assumptions failing makes sense when looking back at our bivariate relationships. All variables in our dataset show heteroskedasticity with expenditure. There are large tight clusters of observations near zero, with the variance widening as the variables increase. All of our data, including expenditure, was heavily right skewed as well. This means that the data itself was full of heteroskedasticity. Because our data contains such a high level of heteroskedasticity throughout, it makes sense that all of our models, even after transformations and higher-order terms, produce residuals that are not normally distributed and do not have constant variance.

Conclusion and Limitation:

Overall, our final model shows a clear improvement over the initial versions and performs well at predicting expenditure. The small difference between the training and testing

adjusted R^2 suggests that the model generalizes effectively, while the low RMSE indicates strong predictive accuracy. These results support the usefulness of the model for estimating expenditure and show that the transformations and interaction terms we introduced had a meaningful impact on performance.

Throughout the modeling process, we identified and addressed several issues in the earlier models, including poor residual behavior and overfitting. While adding transformations and interactions initially improved model fit, it also increased complexity and reduced test performance. By simplifying the model and removing redundant or insignificant terms, we achieved a better balance between accuracy and generalizability. This iterative process helped refine the model and highlighted the importance of model diagnostics and validation.

Despite these improvements, the final model still has limitations. The residuals do not fully meet the assumptions of normality and constant variance, which affects the reliability of standard errors and p-values. As a result, individual predictor interpretations should be treated with caution. However, since the primary goal of this study is prediction rather than inference, these assumption violations have limited impact on the overall usefulness of the model. In summary, the final model is effective for predicting expenditure, but the results should be interpreted with an understanding of the model's assumptions and limitations.

References

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Used to help interpret our model that failed the constant variance assumption.

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Used to help interpret when the normality assumption also fails.